

**SIMATS ENGINEERING**  
**Slot: A, Course Code: ITA0513, Course Name: Computer Vision**

**Duration: 60 mins**

**Marks: 20**

**Assessment Tool 1: Short Answer Test (High)**

**Assessment Tool Weightage: 40%**

**COURSE OUTCOME COVERED**

**CO1:** *Apply the fundamental concepts, image formation models, and processing techniques for analyzing Computer Vision systems. (BTL 3)*

| Q. No. | Questions                                                                                                                                                                  |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | A factory uses cameras to inspect products moving on a conveyor belt. Explain how Computer Vision can identify defective products automatically.                           |
| 2      | A facial recognition system is affected by changes in pose and illumination. Explain which Computer Vision features can help improve recognition performance.              |
| 3      | A traffic monitoring system needs to detect vehicles, recognize traffic signs, and estimate distances. Explain how different levels of Computer Vision are involved.       |
| 4      | A scanned document contains blurred characters and unwanted noise. Explain how digital image fundamentals can be used to improve the document before further processing.   |
| 5      | A smartphone camera produces different image quality under indoor and outdoor conditions. Explain the factors involved in image sensing and acquisition.                   |
| 6      | A surveillance camera captures a moving person as a blurred object. Identify possible causes related to image acquisition and suggest suitable improvements.               |
| 7      | A digital image contains very few pixels and important object details are missing. Explain how the sampling process affects the image quality.                             |
| 8      | An image contains smooth intensity variations, but after digitization it shows noticeable intensity steps. Explain how quantization causes this effect.                    |
| 9      | A remote sensing system must capture very small objects on the Earth's surface. Explain how the choice of sampling resolution affects the ability to detect these objects. |
| 10     | A grayscale image is stored using 2 bits per pixel instead of 8 bits per pixel. Explain how this change affects the number of intensity levels and image appearance.       |

**Code of Conduct:**

*I Keerthana G (192521446) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

**Signature of the Student**  
**Keerthana G**  
**192521446**

**RUBRICS FOR CALCULATION**

| Criteria                  | Weightage | Level 4 (Exemplary)                                                         | Level 3 (Proficient)                              | Level 2 (Developing)                                        | Level 1 (Beginning)                     |
|---------------------------|-----------|-----------------------------------------------------------------------------|---------------------------------------------------|-------------------------------------------------------------|-----------------------------------------|
| Concept Identification    | 25%       | Identifies all key concepts from literature accurately and comprehensively  | Identifies most key concepts with minor omissions | Identifies limited concepts; some are irrelevant or missing | Fails to identify essential concepts    |
| Linkages                  | 25%       | Establishes clear, meaningful, and accurate relationships between concepts  | Shows correct relationships with minor gaps       | Relationships are weak, unclear, or partially incorrect     | No meaningful relationships established |
| Organization              | 20%       | Concepts are well-structured hierarchically with logical flow and grouping  | Mostly organized with minor structural issues     | Some organization but lacks clear hierarchy                 | No logical organization or structure    |
| Integration of Literature | 20%       | Integrates multiple studies effectively to show connections and comparisons | Shows some integration of literature              | Limited integration; mostly isolated concepts               | No integration of literature evidence   |
| Clarity                   | 10%       | Concept map is clear, neat, visually structured, and easy to interpret      | Generally clear with minor readability issues     | Clarity is reduced; difficult to interpret in parts         | Poorly presented and hard to understand |



1. Defective product detection :

- \* Computer vision uses camera to capture product images on a conveyor belt.
- \* Image processing detects features such as
  - \* shape
  - \* colour
  - \* edges
  - \* texture.
- \* The system compares them with normal products and automatically identifies defective products

2. Facial Recognition :

Features such as edges, texture and facial landmarks helps recognize faces under different condition. Image normalization and contrast enhancement can reduce the effect of illumination and pose changes.



3.

### Traffic Monitoring

- \* Low level : filtering enhancement and edge detection.
- \* Mid level : segmentation , features extraction and object tracking
- \* High level : vehicle / sign recognition and distance estimation

4.

### Blurred and Noisy Document

- \* Filtering can remove unwanted noise , while sharpening improves blurred characters.
- \* Contrast Enhancement and thresholding can make the character as makes it clearer for further processing such as OCR.



5.

## Image Sensing and Acquisition

Image quality depends on lighting, camera sensor, lens, exposure, shutter speed, ISO and resolution. Indoor and outdoor lightening conditions change these factors in different image quality

6.

## Blurred Moving person

- \* The Blur is mainly caused by motion during a long exposure time.
- \* It can be reduced by using a faster shutter speed, better lighting and image stabilization.

7.

## Sampling and Image Quality

- \* Sampling converts a continuous image into pixels.
- \* Low sampling resolution produces fewer pixels, causing loss of details and pixelation



- \* Higher sampling resolution represents more spatial details.

8.

### Quantization :

- \* Quantization converts continuous intensity values into a limited number of levels
- \* if fewer levels are used, smooth intensity changes become visible steps or bands causing false contouring

9.

### Sampling Resolution :

- \* Higher spatial resolution provides smaller pixels, allowing small object to be detected more clearly.
- \* Low resolution uses large pixels, so small objects may be lost or mixed.



2-bit vs 8-bit image.

The number of intensity levels is  $2^n$

$$\dagger \quad 2\text{-bit} : 2^2 = 4 \text{ levels}$$

$$\dagger \quad 8\text{-bit} : 2^8 = 256 \text{ levels.}$$

Thus, 2 bit image have fewer shades and more visible intensity steps, while 8-bit images provide smoother grayscale details.